

AWS Documentation

If you're preparing AWS documentation for interviews, projects, or learning, here is a concise **AWS Documentation Content** covering the most commonly used services.

AWS (Amazon Web Services)

AWS is a cloud computing platform that provides on-demand services such as computing power, storage, databases, networking, security, and machine learning. It helps organizations deploy applications without managing physical infrastructure.

1. EC2 (Elastic Compute Cloud)

- Provides virtual servers in the cloud.
- Used to host web applications and APIs.
- Supports Linux and Windows operating systems.
- Allows scaling resources based on demand.

2. S3 (Simple Storage Service)

- Object storage service for storing files, images, videos, and backups.
- Highly durable and scalable.
- Supports versioning and lifecycle management.

3. RDS (Relational Database Service)

- Managed database service.
- Supports MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server.
- Handles backups, patching, and maintenance automatically.

4. Lambda

- Serverless computing service.
- Runs code without managing servers.
- Triggered by events such as API calls, file uploads, or database changes.

5. IAM (Identity and Access Management)

- Manages users, groups, and permissions.
- Provides secure access control for AWS resources.
- Supports roles and policies.

6. VPC (Virtual Private Cloud)

- Creates isolated cloud networks.
- Controls IP ranges, subnets, route tables, and security settings.

- Provides secure communication between resources.

7. CloudWatch

- Monitoring and logging service.
- Tracks application performance and resource utilization.
- Generates alerts based on predefined metrics.

8. CloudFormation

- Infrastructure as Code (IaC) service.
- Automates AWS resource deployment using templates.
- Ensures consistent infrastructure setup.

9. ECS & EKS

- ECS: Container orchestration service.
- EKS: Managed Kubernetes service.
- Used for deploying and managing containerized applications.

10. Route 53

- AWS DNS and domain management service.
- Routes traffic to applications.
- Supports health checks and load balancing.

11. ELB (Elastic Load Balancer)

- Distributes incoming traffic across multiple servers.
- Improves application availability and fault tolerance.

12. Auto Scaling

- Automatically adjusts EC2 instances based on traffic.
- Optimizes performance and cost.

AWS Services Used in Your Projects

Based on your experience:

- **AWS EC2** – Hosting Django applications.
- **AWS RDS** – Managing MySQL databases.
- **AWS Lambda** – Running serverless functions.
- **AWS S3** – Storing project files and backups.
- **IAM** – User and permission management.
- **CloudWatch** – Monitoring application logs and performance.

AWS Benefits

- Scalability
- High Availability
- Security
- Cost Optimization
- Global Infrastructure
- Pay-as-you-go Pricing

Conclusion

AWS provides a comprehensive cloud platform that enables developers to build, deploy, and scale applications efficiently. Services such as EC2, S3, RDS, Lambda, IAM, and CloudWatch help organizations reduce infrastructure management and focus on application development.

This content is suitable for **project documentation, interview preparation, resume explanation, and internship reports.**